J Appl Physiol 108: 866–874, 2010.
First published January 21, 2010; doi:10.1152/japplphysiol.91425.2008.

Effect of recovery duration from prior exhaustive exercise on the parameters of the power-duration relationship

C. Ferguson,[1,2] H. B. Rossiter,[1] B. J. Whipp,[1] A. J. Cathcart,[2] S. R. Murgatroyd,[1] and S. A. Ward[1]

[1]*Institute of Membrane and Systems Biology, Faculty of Biological Sciences, University of Leeds, Leeds; and* [2]*Department of Integrative and Systems Biology, Faculty of Biomedical and Life Sciences, University of Glasgow, Glasgow, United Kingdom*

Submitted 29 October 2008; accepted in final form 14 January 2010

Ferguson C, Rossiter HB, Whipp BJ, Cathcart AJ, Murgatroyd SR, Ward SA. Effect of recovery duration from prior exhaustive exercise on the parameters of the power-duration relationship. *J Appl Physiol* 108: 866–874, 2010. First published January 21, 2010; doi:10.1152/japplphysiol.91425.2008.—The physiological equivalents of the curvature constant (W′) of the high-intensity power-duration (P-t_{LIM}) relationship are poorly understood, although they are presumed to reach maxima/minima at exhaustion. In an attempt to improve our understanding of the determinants of W′, we therefore aimed to determine its recovery kinetics following exhaustive exercise (which depletes W′) concomitantly with those of O_2 uptake ($\dot{V}O_2$, a proxy for the kinetics of phosphocreatine replenishment) and blood lactate concentration ($[L^-]$). Six men performed cycle-ergometer exercise to t_{LIM}: a ramp and four constant-load tests, at different work rates, for estimation of lactate threshold, W′, critical power (CP), and maximum $\dot{V}O_2$. Three further exhausting tests were performed at different work rates, each preceded by an exhausting "conditioning" bout, with intervening recoveries of 2, 6, and 15 min. Neither prior exhaustion nor recovery duration altered $\dot{V}O_2$ or $[L^-]$ at t_{LIM}. Post-conditioning, the P-t_{LIM} relationship remained well characterized by a hyperbola, with CP unchanged. However, W′ recovered to 37 ± 5, 65 ± 6, and 86 ± 4% of control following 2, 6, and 15 min of intervening recovery, respectively. The W′ recovery was curvilinear [interpolated half time ($t_{1/2}$) = 234 ± 32 s] and appreciably slower than $\dot{V}O_2$ recovery ($t_{1/2}$ = 74 ± 2 s) but faster than $[L^-]$ recovery ($t_{1/2}$ = 1,366 ± 799 s). This suggests that W′ determines supra-CP exercise tolerance, its restitution kinetics are not a unique function of phosphocreatine concentration or arterial $[L^-]$, and it is unlikely to simply reflect a finite energy store that becomes depleted at t_{LIM}.

critical power; curvature constant; lactate clearance; oxygen uptake

THE TOLERABLE DURATION (t_{LiM}) of high-intensity muscular exercise has been well described as a hyperbolic function of the external power (P), which asymptotes at what is termed the "fatigue threshold" or "critical power" (CP). The curvature constant (W′) of the hyperbola is mathematically equivalent to a constant amount of work that can be performed above CP, i.e., the product of P and t_{LIM} (e.g., 27, 41, 42, 45, 60). CP is generally agreed to represent the highest sustainable work rate (WR) for which steady states of pulmonary O_2 uptake ($\dot{V}O_2$) and the elevated blood lactate ($[L^-]$) and proton ($[H^+]$) concentration responses can be achieved (e.g., 29, 45, 59), although some investigators have reported CP to exceed slightly the WR at the "maximum lactate steady state" (e.g., 26, 36, 47). Hence, CP demarcates the transition between what has been termed heavy-intensity and very heavy-intensity exercise (59). However, the physiological basis of W′ remains controversial.

W′ is defined through the relationship between P and t_{LIM}, in this sense it is itself a "simple" mathematical value. However, its value presumably reflects some physiological variable (or variables) that (with CP) determines t_{LIM}. It is commonly suggested that the physiological equivalents of W′ are represented by a finite energy store comprising intramuscular high-energy phosphates [ATP and phosphocreatine (PCr)], glycogen, and stored O_2 (14, 38–40, 42, 45, 46). This coheres with the suggestion by some investigators that W′ is synonymous with the maximum O_2 deficit (24) or anaerobic work capacity (23, 42; although see Ref. 61) and that it is depleted at a rate that bears some proportionality to the magnitude of the power requirement above CP, with exhaustion occurring when this apparent store is depleted (42, 45). More recently, however, W′ has been proposed to reflect the build-up of fatigue-inducing metabolites (e.g., the intramuscular accumulation of inorganic phosphate and H^+ and the interstitial/extracellular accumulation of K^+) to some critical tolerable limit (7, 13, 28, 45). This, in turn, would require the rate of metabolite build-up to be proportionally coupled to the rate of W′ "utilization" (7).

Establishing the physiological correlate(s) of W′ is complicated by the difficulty in resolving whether their profiles of change during exercise (be it depletion or accumulation) are reflected in the rate of utilization of W′. It seems likely that the putative physiological equivalents would reach some minimum (or maximum) value at the point at which exercise intolerance is reached, with W′ then being mathematically defined. However, whether a predictable relationship [either linear or some more complex function (e.g., 46)] between W′ at its equivalents during exercise can be made requires knowledge of the physiological underpinnings of the W′ construct. That is, the relationship between W′ and its putative physiological determinants might be established by independent experimental manipulation of W′, with CP remaining constant. Here we propose a strategy to achieve this based on our recent demonstration that W′ was reduced following a conditioning bout of supra-CP exercise with an intervening 2-min recovery, whereas key parameters of aerobic function (e.g., the fundamental $\dot{V}O_2$ time constant, $\dot{V}O_{2\,max}$, and CP) were unaffected (10). Consequently, redefining the P-t_{LIM} relationship after exhaustive supra-CP exercise with a range of intervening recovery durations may provide a means of establishing the recovery kinetics of W′ in concert with those of its putative physiological determinants. It is suggested that, by establishing the kinetics of W′ and its physiological equivalents (rather than simply a single W′ value at the point of intolerance, as has been the case to date), insight would be provided into the determinants of exercise tolerance.

Address for reprint requests and other correspondence: C. Ferguson, Dept. of Integrative and Systems Biology, Faculty of Biomedical and Life Sciences, Univ. of Glasgow, Glasgow G12 8QQ, UK (e-mail: C.Ferguson@bio.gla.ac.uk).

 8750-7587/10 $8.00 Copyright © 2010 the American Physiological Society

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

Consequently, the goal of this study is to determine the kinetics of W′ recovery from exhausting supra-CP constant-load cycle ergometry and to compare them with those of $\dot{V}O_2$ [a reasonable proxy for intramuscular PCr kinetics (e.g., 48)] and arterialized-capillary blood [L^-] (a broad index of recovery arterial lactate clearance). We hypothesized that the degree to which these quantities were kinetically correlated with W′ would allow judgments to be made regarding the involvement of putative energy-store and fatigue-metabolite changes in the restitution and, therefore, composition of W′.

METHODS

Subjects

Six recreationally active, healthy men (24 ± 4.2 yr old, 179.6 ± 7.5 cm height, 86.5 ± 15.3 kg body wt) provided written informed consent to participate in the study. The procedures and protocols were approved by the School of Biomedical Sciences/School of Sport and Exercise Sciences (University of Leeds) Ethical Review Committee and were conducted in accordance with the Declaration of Helsinki.

Equipment

The equipment is described in detail in an earlier report (10). Briefly, subjects exercised on a computer-controlled, electromagnetically braked cycle ergometer (Excalibur Sport, Lode, Groningen, The Netherlands). Ventilatory and pulmonary gas exchange variables were determined breath-by-breath using mass spectrometry (MSX, NSpire, Kent, UK) and turbinometry (Interface Associates, Laguna Niguel, CA). The volume and gas concentration signals were sampled and digitized every 20 ms and time aligned for on-line breath-by-breath measurement of ventilatory and pulmonary gas exchange variables (4). Heart rate was calculated beat-by-beat from the R-R interval of a 12-lead electrocardiogram (Quest, Burdick, WA), and arterial O_2 saturation was monitored noninvasively using finger pulse oximetry (Biox 3745, Ohmeda, Louisville, KY).

Protocols

Subjects were initially familiarized with all protocols and procedures. For each subject, only one exercise test was conducted on a given day (therefore requiring a total of ~14 visits/subject), with each individual participating in no more than three experimental sessions in any given week. Prior to each session, subjects were asked to refrain from participation in strenuous physical activity (preceding 24 h), alcohol ingestion (preceding 48 h), and caffeine ingestion (preceding 3 h) and to arrive at least 2 h postprandial. All testing was commenced from a 20-W baseline (≥4 min) and concluded with a final 20-W recovery phase (≥6 min).

Protocol 1: maximal incremental ramp test. A 25 W/min incremental ramp test was performed to the tolerable limit, defined as the point at which subjects were unable to maintain a cycling cadence of >60 rpm despite encouragement (Fig. 1*A*). Peak $\dot{V}O_2$ ($\dot{V}O_{2\,peak}$) was determined as the average $\dot{V}O_2$ for an integral number of breaths over the final ~20 s of the incremental phase, and the lactate threshold (θ_L) was estimated ($\hat{\theta}_L$) using standard pulmonary gas exchange criteria (62).

Protocol 2: determination of the control P-t_{LIM} relationship. Each subject then completed a randomized series of four constant-load tests to the limit of tolerance, with each test implemented at a different WR chosen to span a t_{LIM} range of ~3–12 min (Fig. 1*B*). CP and W′ were estimated as the power intercept and slope, respectively, of the least-squares linear regression of P vs. $t_{LIM}{}^{-1}$, i.e., $P = (W'/t_{LIM}) + CP$ (23, 45, 60). The standard error (SE) of each individual CP estimation was less than ±3 W and that for W′ was less than ±1.25 kJ. The WR predicted to induce exhaustion at 6 min (WR_6) was

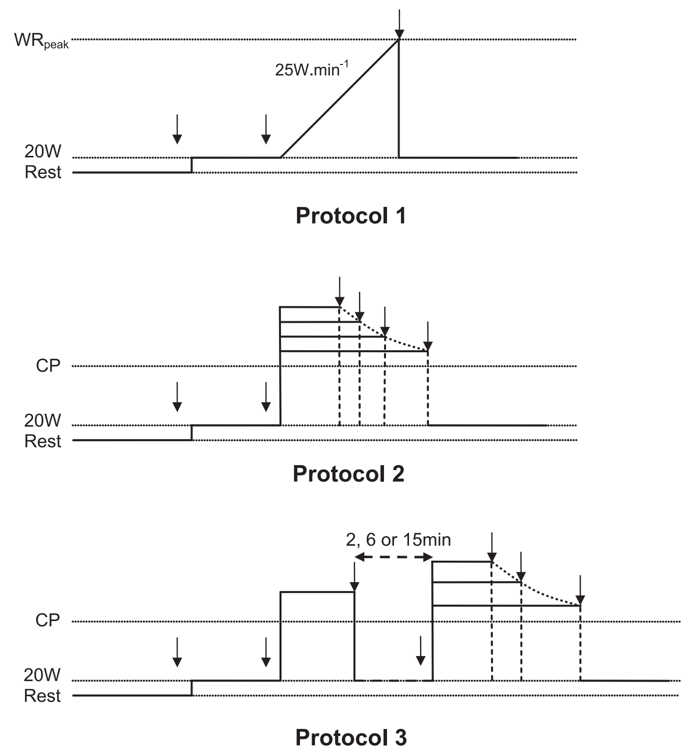

Fig. 1. Schematic representation of the work rate (WR) profiles performed during exercise *Protocols 1* (*A*), *2* (*B*), and *3* (*C*). *Protocol 1*: maximal incremental ramp test. *Protocol 2*: series of 4 constant-load tests, each at different WRs for determination of the power duration (P-t_{LIM}) relationship (dotted line). *Protocol 3*: exhaustive supra-critical power (supra-CP) conditioning exercise [conducted at WR predicted to induce exhaustion at 6 min (WR_6)] followed by a series of 3 constant-load tests, each at different WRs, for determination of the P-t_{LIM} relationship (dotted line). *Protocol 3* was performed with intervening 20-W recoveries of 2, 6, and 15 min. All tests were conducted to the limit of tolerance. Arrows represent blood sampling time points.

derived by interpolation and, subsequently, used as the WR of the exhaustive "conditioning" exercise bout in *protocol 3*.

Protocol 3: effects of recovery duration on the postconditioning P-t_{LIM} relationship. The purpose of this protocol was to estimate the recovery kinetics of W′ concomitantly with those of $\dot{V}O_2$ and arterialized blood [L^-]. The P-t_{LIM} relationship was therefore redefined following a conditioning bout of constant-load exercise performed to the limit of tolerance at WR_6 for each of the three intervening 20-W recovery durations (i.e., 2, 6, and 15 min; Fig. 1*C*). For each recovery condition, subjects completed three supra-CP constant-load exercise tests to the limit of tolerance (cf. *Protocol 2*), where possible at WRs that corresponded to those previously utilized in *Protocol 2*. As a result of postconditioning reductions in W′, on occasion some of the higher WRs performed during the control P-t_{LIM} characterization would have been of too short a duration to be meaningfully used for determination of the postconditioning P-t_{LIM} relationships (Fig. 2). To constrain attainment of t_{LIM} to within the recommended range of ~3–12 min (e.g., 14, 45, 46), in these instances the WRs in question therefore had to be decreased slightly. Tests were randomly assigned.

Blood Sampling

At designated points in all protocols (Fig. 1, arrows), capillary blood samples (~25 μl) were taken from the fingertip of the heated hand and analyzed immediately after the test for [L^-] using an automated analyzer (GM-7, Analox Instruments, London, UK). Before analysis of each set of blood samples, the analyzer was calibrated

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

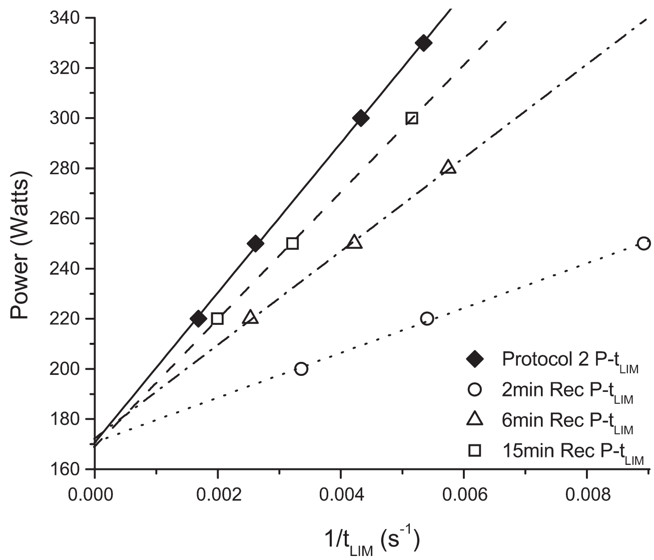

Fig. 2. Linearized P-t_{LIM} relationship for a representative subject under control conditions (*Protocol 2,* solid symbols) and after supra-CP conditioning exercise performed to the limit of tolerance (at WR_6) and intervening 20-W recoveries of 2, 6, and 15 min (*Protocol 3*: open symbols). CP is estimated from the *y*-intercept and the curvature constant (W′) from the slope of the linear regression. CP was consistently unchanged and W′ was consistently reduced (and highly dependent on the intervening 20-W recovery duration) following exhaustive conditioning exercise.

using an 8 mM standard solution, the concentration of which was also checked postanalysis.

Analysis

Each breath-by-breath data set was edited to eliminate any occasional extraneous breaths (more than ±4 SD of the local mean) (34). For control tests (*Protocol 2*), baseline $\dot{V}O_2$ was calculated as the mean $\dot{V}O_2$ over the final 60 s of the 20-W warm-up phase. Following the exhaustive conditioning bout at WR_6, the "baseline" $\dot{V}O_2$ was calculated as the mean $\dot{V}O_2$ recorded in the last 20 s of recovery prior to the imposition of the subsequent supra-CP constant-load test. $\dot{V}O_{2\,peak}$ was calculated as the average $\dot{V}O_2$ for an integral number of breaths over the last ~20 s prior to attaining the limit of tolerance. As $\dot{V}O_{2\,peak}$ was found not to vary over the range of different work rates performed for any subject (see RESULTS), the mean $\dot{V}O_{2\,peak}$ value was taken as $\dot{V}O_{2\,max}$ (8). $\dot{V}O_{2\,max}$ and the duration of the conditioning bout (conducted at WR_6) were each monitored throughout the study to check for the presence of any training effect. These were each found to vary randomly within the day-to-day limits of 10% reported by Day et al. (8) for similarly intense fatiguing constant-load exercise, from which we concluded that our results were not influenced by a training effect.

We did not attempt to formally partition individual $\dot{V}O_2$ responses into their kinetic components (e.g., 28, 44), as the low signal-to-noise ratio for single transitions (because of breath-to-breath variability) (34) and the absence of a discernible steady state or asymptote in $\dot{V}O_2$ precluded an acceptable level of statistical confidence for the parameter estimations. Even single repeats of each of the tests would have required an unacceptably large number of studies (i.e., given the initial ~14 tests typically required of each subject). Not only would this have been overly demanding for the subjects, but it would also be quite likely to introduce a training effect.

Statistical Analysis

The significance of the effect of recovery duration following exhausting exercise at WR_6 on the parameters of the P-t_{LIM} relationship (i.e., CP and W′), $\dot{V}O_2$, and [L^-] were compared using repeated-measures ANOVA and post hoc analysis (Tukey's honestly significant difference) where necessary. Values are means ± SD unless otherwise stated.

RESULTS

Protocol 1: Maximal Incremental Ramp Test

$\dot{V}O_{2\,peak}$ averaged 3.82 ± 0.55 l/min, while $\hat{\theta}_L$ averaged 1.87 ± 0.35 l/min (equivalent to 49 ± 5% $\dot{V}O_{2\,peak}$).

Protocol 2: Control P-t_{LIM} Relationship

The P-t_{LIM} relationship for each subject conformed well to a hyperbola, as demonstrated by the goodness-of-fit of the P-$t_{LIM}{}^{-1}$ relationship to a linear function ($R^2 > 0.996$ in all cases). The SE of each CP estimation was, on average, 1.7 W (range 1.2–2.4 W) and that of W′ was, on average, 0.48 kJ (range 0.34–0.69 kJ; Fig. 2) (cf. Refs. 7, 10). CP and W′ averaged 212 ± 34 W and 21.60 ± 5.2 kJ, respectively (Table 1), and were unaltered when power was set as the independent variable [CP = 212 ± 34 W (P = 0.999), W′ = 21.73 ± 6.26 kJ (P = 0.875)] (cf. Refs. 7, 10). WR_6 interpolated from the P-$t_{LIM}{}^{-1}$ relationship averaged 269 ± 34 W. $\dot{V}O_{2\,peak}$ attained during these constant-load tests was not influenced by WR (P = 0.426), thus meeting the criterion for $\dot{V}O_{2\,max}$, which averaged 3.78 ± 0.56 l/min for the group as a whole (Table 1). [L^-] at the limit of tolerance ([L^-]$_{LIM}$) was also not influenced by WR (P = 0.203) and averaged 10.14 ± 0.97 mM (Table 1), representing an increase (Δ[L^-]) from the preexercise (20-W) baseline of 9.09 ± 1.16 mM.

Table 1. *Group mean values of maximum O_2 uptake, CP, W′, and [L^-] at [L^-]$_{LIM}$ in protocols 2 and 3*

		Protocol 3		
	Protocol 2	2 min Rec	6 min Rec	15 min Rec
Maximum O_2 uptake, l/min	3.78 ± 0.56	3.64 ± 0.50	3.76 ± 0.50	3.73 ± 0.49
CP, W	212 ± 34	213 ± 36	213 ± 34	213 ± 36
SE of CP estimate, W	1.7 (1.2–2.4)	1.1 (0.5–2.1)	1.1 (0.1–2.3)	2.1 (0.7–3.1)
W′, kJ	21.60 ± 5.16	7.8 ± 1.40*	14.1 ± 3.70*†	18.5 ± 4.60*†‡
SE of W′ estimate, kJ	0.48 (0.34–0.69)	0.20 (0.01–0.33)	0.26 (0.01–0.54)	0.72 (0.15–1.21)
[L^-]$_{LIM}$, mM	10.14 ± 0.97	10.38 ± 1.34	10.77 ± 1.15	10.23 ± 1.31

Values are means ± SD, except critical power (CP) and curvature constant (W′) estimates, which are shown as mean SE and range. [L^-]$_{LIM}$, lactate concentration measured at the limit of tolerance. *Protocol 2*, control values; *Protocol 3*, values obtained after exhaustive conditioning exercise at work rate predicted to induce exhaustion at 6 min (WR_6) with intervening 20-W recoveries of 2, 6, and 15 min (2 min Rec, 6 min Rec, and 15 min Rec). *Significantly different from *Protocol 2* ($P < 0.05$). †Significantly different from 2 min Rec ($P < 0.05$). ‡Significantly different from 6 min Rec ($P < 0.05$).

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

Protocol 3: Effects of Recovery Duration on the Postconditioning P-t_{LIM} Relationship

The tolerable duration of the WR_6 conditioning bout averaged 366 ± 21 s, which was not significantly different between subjects ($P = 0.537$) or between each of the nine bouts within subjects ($P = 0.635$). $\dot{V}O_{2\,peak}$ (3.81 ± 0.52 l/min) in the conditioning bout was not significantly different from $\dot{V}O_{2\,max}$ ($P = 0.969$). $[L^-]_{LIM}$ was also not significantly different between each of the nine conditioning bouts ($P = 0.884$) or from the control value obtained during *Protocol 2* (i.e., 10.14 ± 0.97 mM, $P = 0.288$), averaging 10.11 ± 1.02 mM (Table 2), which was equivalent to $\Delta[L^-]$ of 9.06 ± 1.16 mM.

Following the supra-CP conditioning bout, t_{LIM} at any particular WR was systematically reduced compared with control (*Protocol 2*) and to an extent that was more marked the shorter the intervening 20-W recovery duration, averaging 38 ± 8, 65 ± 5, and 84 ± 5% of the control duration following intervening 20-W recoveries of 2, 6, and 15 min, respectively. The P-t_{LIM} relationship remained hyperbolic, with the R^2 of the P-t_{LIM}^{-1} linear transform being >0.994 and the SE of CP and W′ estimates remaining within ±3 W and 1.25 kJ, respectively (see Table 1 for mean and range of values). CP was unchanged from control values (*Protocol 2*, 212 ± 34 W), regardless of the intervening recovery duration ($P = 0.922$), averaging 213 ± 36, 213 ± 34, and 213 ± 36 W following 20-W recoveries of 2, 6, and 15 min, respectively (Figs. 2 and 3, Table 1).

In contrast, compared with control values, W′ was significantly reduced ($P = 0.001$) in each of the postconditioning cases by an amount that was highly dependent on the intervening recovery duration (Figs. 2 and 3). Following the 2-min 20-W recovery, W′ averaged 7.8 ± 1.4 kJ, equivalent to a 37 ± 5% recovery, which was significantly lower than the control estimate (*Protocol 2*, $P = 0.001$). The 6-min recovery resulted in a greater repletion of W′, to 14.1 ± 3.7 kJ (corresponding to 65 ± 6% recovery). However, although this value attained at 6 min was significantly greater than that obtained following the 2-min recovery ($P = 0.002$), W′ remained significantly lower than control (*Protocol 2*, $P = 0.001$). Following the 15-min recovery, W′ averaged 18.5 ± 4.6 kJ, equivalent to 86 ± 4% repletion. However, although this is significantly greater than the level to which W′ recovered following both the 2-min ($P = 0.001$) and 6-min ($P = 0.001$) recoveries, it was still significantly lower than the control estimate (*Protocol 2*, $P = 0.001$; Table 1).

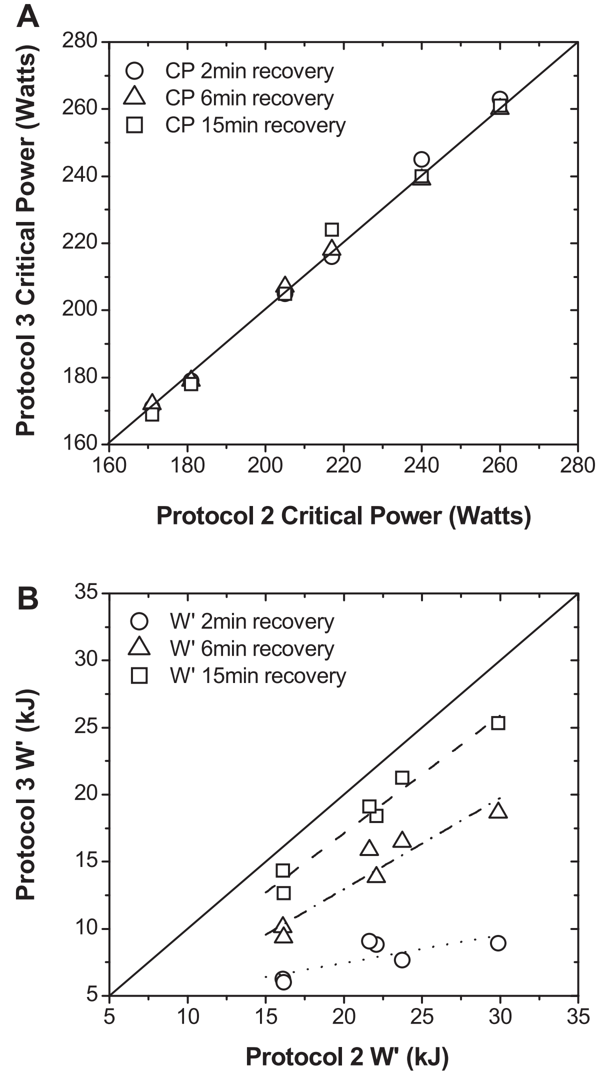

Fig. 3. CP (*A*) and W′ (*B*) under control conditions (*Protocol 2*) compared with estimates obtained following exhaustive conditioning exercise and intervening 20-W recoveries of 2, 6, and 15 min (*Protocol 3*). Individual values for each subject are compared against line of identity. Note that CP was unchanged following supra-CP conditioning exercise, conducted to the tolerable limit, regardless of the intervening recovery duration, while W′ was consistently reduced and highly dependent on the intervening recovery duration.

The baseline value to which $\dot{V}O_2$ recovered following the conditioning bout at WR_6 (i.e., prior to the onset of the postconditioning supra-CP bouts) was significantly dependent on the intervening recovery duration ($P = 0.001$; Fig. 4). That is, following the 2-min 20-W recovery, $\dot{V}O_2$ averaged 1.37 ± 0.20 l/min, which was significantly higher than the control 20-W baseline obtained prior to the onset of WR_6 (0.74 ± 0.08 l/min, $P = 0.001$). There was a further significant, although still incomplete, recovery of $\dot{V}O_2$ after the 6-min 20-W recovery period (i.e., to 1.06 ± 0.12 l/min) compared with the 2-min 20-W recovery ($P = 0.001$). $\dot{V}O_2$ also continued to recover throughout the 15-min 20-W recovery phase to values significantly lower than those obtained during both the 2-min ($P = 0.001$) and 6-min ($P = 0.001$) recovery phases (i.e., to 0.87 ±

Table 2. *Blood $[L^-]$ at the limit of tolerance of the exhaustive conditioning bout (WR_6) and after 2, 6, and 15 min of 20-W recovery*

	$[L^-]$, mM			
		Postconditioning recovery		
Subj No.	WR_6	2 min Rec	6 min Rec	15 min Rec
1	11.25	10.00	10.35	8.30
2	10.09	9.90	8.07	6.93
3	10.61	11.15	9.37	7.27
4	9.11	10.08	9.00	4.90
5	8.69	8.40	7.23	4.13
6	10.93	10.48	10.50	7.03
Mean ± SD	10.11 ± 1.02	10.00 ± 0.91	9.09 ± 1.28	6.43 ± 1.58*†

WR_6, limit of tolerance of exhaustive conditioning bout; $[L^-]$, blood lactate concentration. *Significantly different from WR_6 ($P < 0.05$). †Significantly different from 6 min Rec ($P < 0.05$).

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

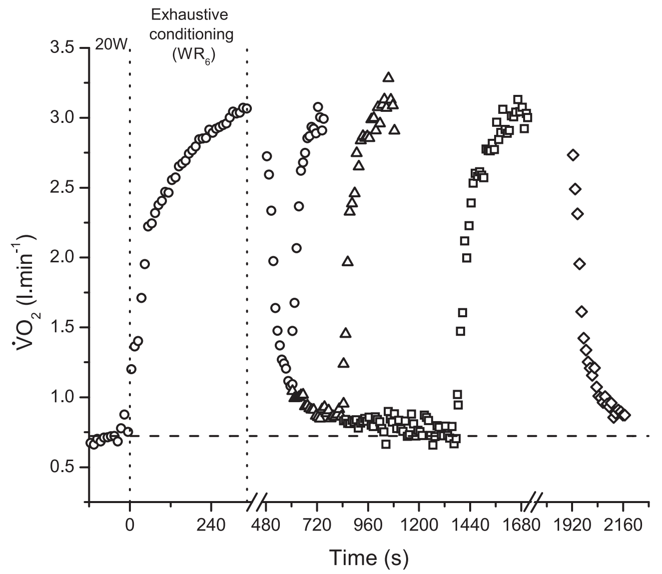

Fig. 4. O_2 uptake ($\dot{V}O_2$) response for a representative subject during exhausting conditioning exercise and during subsequent supra-CP exercise performed at the same WR following 2 min (○), 6 min (△), and 15 min (□) of 20-W recovery. Note the continued recovery of $\dot{V}O_2$ as the intervening recovery duration increased. Also note the constancy of peak $\dot{V}O_2$ ($\dot{V}O_{2\,peak}$), independent of recovery duration. $\dot{V}O_2$ recovery from the limit of tolerance in the 2nd bout is shown as the average of all bouts (◇).

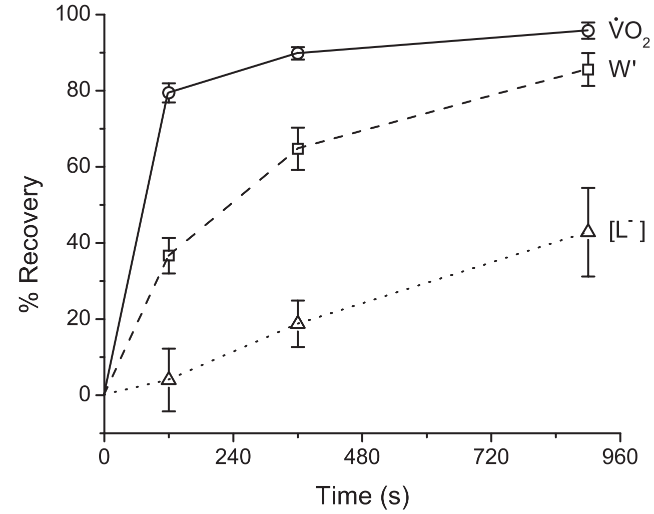

Fig. 5. Mean temporal profile of the magnitude of W′ recovery (□) relative to recovery of $\dot{V}O_2$ (○) and lactate concentration ([L⁻], △); bars represent SD. Note faster recovery kinetics of $\dot{V}O_2$ than of W′ and slower recovery kinetics of [L⁻] than of W′.

0.13 l/min) but remained significantly higher ($P = 0.017$) than the preexercise baseline (Table 3). However, the profile of $\dot{V}O_2$ recovery was appreciably faster than that of W′, as evident from the linearly interpolated time profiles (Fig. 5), having an interpolated half time ($t_{1/2}$) of 74 ± 2 s compared with 234 ± 32 s for W′. This discrepancy between the recovery kinetics of $\dot{V}O_2$ and W′ is also evident in the proportionality of their recoveries (Fig. 6*A*), with a given %$\dot{V}O_2$ recovery associated with a much smaller %W′ recovery.

Similarly, the degree to which [L⁻] recovered following exhaustive exercise at WR_6 was significantly dependent on recovery duration. There was no significant decrease in [L⁻] from the end-exercise WR_6 value after 2 min of recovery at 20 W ($P = 0.990$), with [L⁻] averaging 10.00 ± 0.91 mM prior to the onset of the postconditioning tests. After the 6-min 20-W recovery, [L⁻] had fallen (to 9.09 ± 1.28 mM) relative to the end-exercise WR_6 value, although this was not statistically significant ($P = 0.061$). The 15-min recovery period allowed a further decrease in [L⁻] compared with the end-exercise WR_6 value ($P = 0.001$), averaging 6.43 ± 1.58 mM prior to the onset of the second phase of the protocol (Table 2). However, even following the 15-min 20-W recovery, [L⁻] remained significantly elevated compared with the control 20-W baseline value ($P = 0.001$; Figs. 6*B* and 7, Table 2). Furthermore, in contrast to $\dot{V}O_2$, recovery of [L⁻] was slower than that of W′ (Fig. 5), with an interpolated $t_{1/2}$ = 1,366 ± 799 s (because of the slow [L⁻] recovery relative to the recovery time frame, for simplicity, the $t_{1/2}$ estimation assumed that the immediately subsequent [L⁻] recovery followed the same profile that was observed between 6 and 15 min). This temporal discrepancy is evident in the proportional recoveries of W′ and [L⁻] (Fig. 6*B*).

Table 3. *"Baseline" O_2 uptake immediately prior to determination of the P-t_{LIM} relationship in Protocols 2 and 3*

	O₂ Uptake, l/min			
		Protocol 3		
Subj No.	*Protocol 2*	2 min Rec	6 min Rec	15 min Rec
1	0.68	1.13	0.87	0.71
2	0.73	1.30	1.04	0.93
3	0.62	1.32	1.05	0.75
4	0.82	1.63	1.21	1.03
5	0.74	1.26	1.00	0.82
6	0.84	1.59	1.16	0.99
Mean ± SD	0.74 ± 0.08	1.37 ± 0.20*	1.06 ± 0.12*†	0.87 ± 0.13*†‡

P, external power; t_{LIM}, tolerable duration of high-intensity muscular exercise. *Significantly different from *Protocol 2* ($P < 0.05$). †Significantly different from 2 min Rec ($P < 0.05$). ‡Significantly different from 6 min Rec ($P < 0.05$).

$\dot{V}O_{2\,peak}$ at the end of *Protocol 3* (i.e., at the limit of tolerance in the 2nd bout of *Protocol 3*) was not influenced by WR following either the 2-min ($P = 0.053$), 6-min ($P = 0.436$), or 15-min ($P = 0.071$) recovery (again meeting the $\dot{V}O_{2\,max}$ criterion; see Table 1 for $\dot{V}O_{2\,max}$ values). Furthermore, postconditioning $\dot{V}O_{2\,max}$ was invariant across each of the recovery durations and corresponded to $\dot{V}O_{2\,max}$ attained from control supra-CP constant-load tests (*Protocol 2*, $P > 0.05$; Fig. 4). $\dot{V}O_2$ attained at the limit of tolerance following 2 min of recovery was 0.17 ± 0.16 l/min lower than that attained during the WR_6 conditioning bout ($P = 0.015$) but remained well within the expected test-retest variability of ~10% (8). Similarly, $[L^-]_{LIM}$, attained following intolerance in either the conditioning bout or subsequent bouts, was similar between all protocols (Table 1, Fig. 7).

DISCUSSION

This, we believe, is the first study to attempt characterization of W′ recovery kinetics and its putative physiological equivalents for supra-CP exercise. The data demonstrate that W′ reconstitution following exercise is curvilinear (cf. Ref. 46), as determined by supra-CP exercise to t_{LIM} after an exhausting conditioning bout and subsequent variable recovery periods,

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

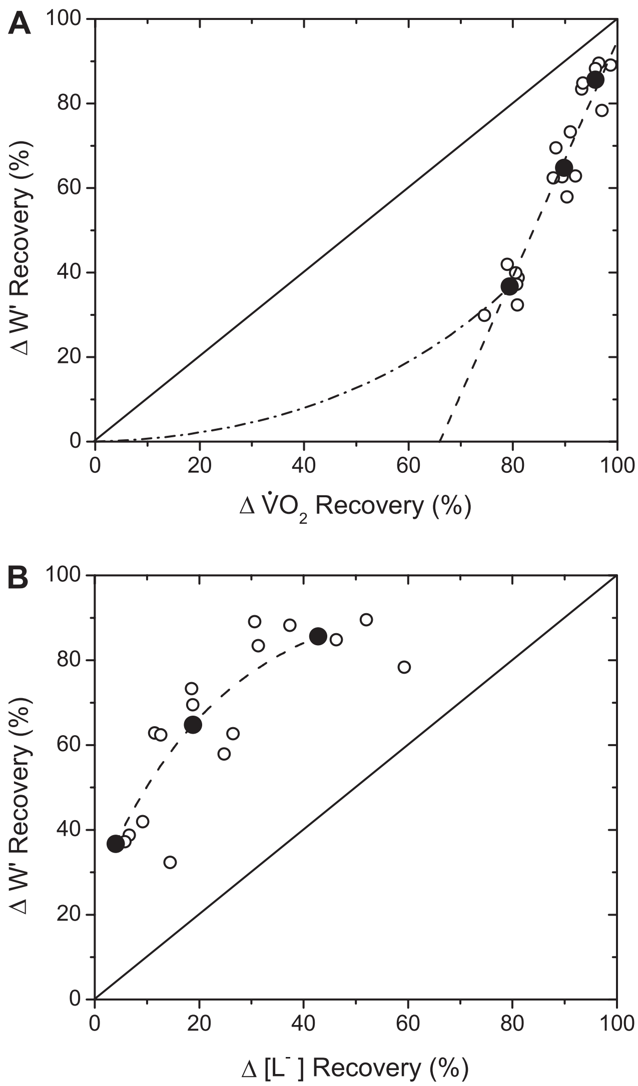

Fig. 6. Proportional relationship between the magnitude of W′ recovery and recovery of $\dot{V}O_2$ (*A*) and $[L^-]$ (*B*), expressed as percentages of the corresponding control responses. Individual (○) and mean (●) responses are plotted for each recovery duration (i.e., 2, 6, and 15 min). For $\dot{V}O_2$, best fits to a simple linear function (dashed line) and to a curvilinear function projecting to the origin (dashed-dotted line) are shown. For $[L^-]$, best-fit curvilinear function is shown. Solid lines projecting though the origin are lines of identity.

being progressively reduced the shorter the intervening recovery (20-W) period, while CP remained unchanged. The constancy of CP means that the reduction in t_{LIM} following the exhaustive conditioning exercise with variable recovery durations (2–15 min) is solely dependent on W′. This extends our previous work (10), which used a single 2-min recovery period, and suggests that W′ reconstitution is not linear (as inferred from its mathematical calculation) but follows a trajectory that lies somewhere between the physiological surrogates for the "depletion" and "accumulation" hypotheses measured here: neither $\dot{V}O_2$ nor $[L^-]$ recovery well represented the W′ profile. However, given that W′ appears only to be replenished during exercise if recovery WRs are below CP [allowing predominantly aerobic energy transfer (7, 14)], these observations together demonstrate that W′ depletion alone shapes the tolerance of supra-CP exercise, thus justifying preliminary attempts to gain insight into its physiological determinants.

That exhaustive supra-CP conditioning exercise did not affect CP or, indeed, $\dot{V}O_{2\,max}$, is consistent with our earlier findings (10) and those from others using heavy-intensity (28) and sprint (22, 58) conditioning. Thus W′ depletion during the conditioning bout does not appear to influence CP, in contrast to previous suggestions (7). Although this supports the "anaerobic" constitution of W′, it does not rule out W′ being related to aerobic energetic parameters via its influence on the energetic equivalency of CP; that is the rate of ATP turnover to sustain CP may not be constant. For example, W′ depletion by prior supra-CP exercise has been reported to reduce work efficiency (52) and, as such, may play a role in increasing the aerobic metabolic equivalent of CP and, thus, limiting the "role" of W′ in inducing intolerance in a subsequent bout. Therefore, W′ might better reflect influences related to fatigue induction, rather than those solely of anaerobic energy-transfer origin.

We suggested that insight into the physiological determinants of W′ might be derived from its temporal recovery profile relative to those of $\dot{V}O_2$ [a reasonable proxy for intramuscular PCr kinetics (48)] and $[L^-]$ (a broad, albeit coarse, index of blood lactate clearance under these conditions) following exhausting supra-CP conditioning exercise. While these profiles were each clearly curvilinear (Fig. 5), we did not attempt formal kinetic characterization (e.g., exponential, hyperbolic, or some more complex function), given the small number of data points in each subject data set ($n = 3$). Rather, we confined ourselves to simply estimating the interpolated $t_{1/2}$ and inferring what these corelationships may suggest regarding the constitution of W′ with respect to depletion of putative substrate pools and/or accumulation of key fatigue-inducing metabolites.

Phosphocreatine

Intramuscular PCr stores have been proposed to be one of the major determinants of W′ (14, 38, 40, 42, 45, 46, 55). The

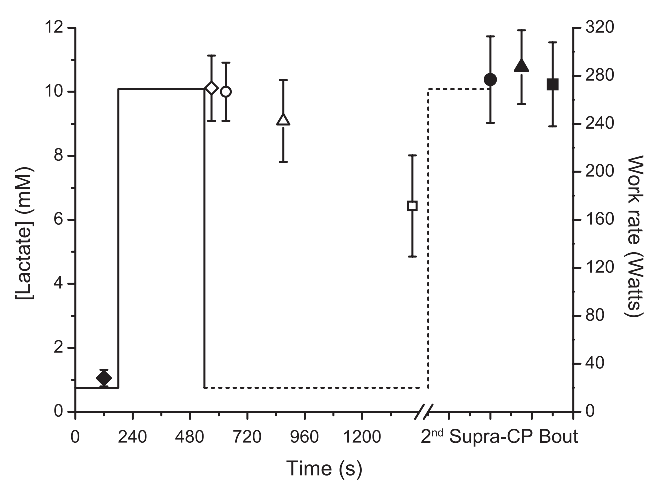

Fig. 7. $[L^-]$ measured throughout *Protocol 3*. $[L^-]$ was measured during the initial warm-up (◆), at the limit of tolerance of the exhaustive conditioning bout (at WR_6; ◇), and at the termination of intervening 20-W recoveries of 2 min (○), 6 min (△), and 15 min (□). $[L^-]$ was also measured at the limit of tolerance of subsequent constant-load exercise ($[L^-]_{LIM}$) following intervening recoveries of 2 min (●), 6 min (▲), and 15 min (■). Mean responses are plotted along with error bars (SD). Note the constancy of$[L^-]_{LIM}$, independent of prior exhaustive exercise and the intervening 20-W recovery duration.

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

classical characterization of intramuscular PCr concentration ([PCr]) kinetics following fatiguing exercise in terms of parallel "fast" and "slow" components (18) coheres with the more recent description of $\dot{V}O_2$ recovery kinetics from very heavy-intensity exercise in terms of "fundamental" and "slow" component elements (44). Furthermore, [PCr] kinetics both during and following high-intensity exercise have been shown to be closely correlated with those of $\dot{V}O_2$ (e.g., 48), the later providing a close estimate of muscle O_2 consumption kinetics (2, 17). It might therefore be reasonably supposed that the $\dot{V}O_2$ recovery profile following supra-CP conditioning was reasonably reflective of the corresponding intramuscular [PCr] profile. However, it should be noted that it is only the fraction of the entire intramuscular PCr pool actually expended above CP that would be involved in W′ replenishment in the subsequent recovery. Indeed, while the fractional recoveries of W′ and $\dot{V}O_2$ were linearly related following fatiguing exercise, W′ clearly recovered more slowly than $\dot{V}O_2$ ($t_{1/2}$ = 234 ± 32 vs. 74 ± 2 s; Figs. 5 and 6*A*).

W′ recovery relative to that of $\dot{V}O_2$ recovery was relatively linear over the range investigated (Fig. 6*A*). However, its linear extrapolation to the $\dot{V}O_2$ axis suggests that, following the attainment of $\dot{V}O_{2\,max}$ at t_{LIM}, no W′ recovery could occur until $\dot{V}O_2$ had recovered by some 60%. We consider this to be an unlikely scenario, as W′ is known to recover whenever WR falls below CP (7, 14). Also, this extrapolated value is approximately equal to the group mean θ_L (1.87 ± 0.35 l/min). A more curvilinear W′ recovery profile (Fig. 6*A*, dashed-dotted line) thus seems more likely, although the precise relationship is an issue requiring resolution. Regardless, the discrepancy between the W′ and $\dot{V}O_2$ recovery kinetics seems to suggest that if PCr is indeed a determinant of W′, then other mechanisms with slower recovery kinetics also contribute.

Muscle Glycogen

As one such source has been proposed to be the intramuscular glycogen store (14, 40, 42, 45, 46), it would be of interest to establish whether glycogen loading can increase W′. Glycogen depletion has been shown to decrease W′ (39), although *1*) an ~50–70% decrease in muscle glycogen (20) has been demonstrated to result in only an average 20% decrease in W′ (39) and *2*) the global muscle glycogen content is unlikely to be limiting over the exercise durations we utilized (53). However, the heterogeneous properties and recruitment patterns of muscle raise the possibility that a degree of fiber-type dependent regional glycogen depletion (16, 57) may account for (or contribute to) the ~15% shortfall in W′ reconstitution that was evident by the end of the 15-min recovery (Fig. 5). The exact mechanism(s) through which such glycogen depletion might be expressed is uncertain but may include *1*) a reduced pyruvate availability that constrains mitochondrial respiration rate, independently of any effect on pyruvate dehydrogenase (37) and of anaplerotic reactions leading to falls in key citric acid cycle intermediates (e.g., 51), and *2*) impaired excitation-contraction coupling consequent to factors such as reduced sarcoplasmic reticulum Ca^{2+} release (1). Regardless, the large dissociation between the decrease in muscle glycogen (20) and the decrease in W′ (39), coupled with evidence suggesting that intramuscular ATP concentration is unchanged (e.g., 48) and the total intramuscular PCr store is not completely depleted at t_{LIM} (e.g., 29, 31, 49) (although PCr depletion in individual fibers, reflecting intramuscular heterogeneities, is of course possible), suggests that it is unlikely that W′ is simply reflecting a source of stored energy that is depleted during supra-CP exercise.

Lactic Acidosis

The role of the fatigue mediators L^- (e.g., 9, 25) and H^+ (e.g., 11) in W′ repletion kinetics is unclear, as we were not able to monitor intramuscular levels and regional distributions of L^- and H^+. Studies from the literature suggest that muscle [L^-] is likely still to be falling at the 15-min recovery point (e.g., 21, 50), although the constraints on biopsy sampling limit descriptions to only a few points dispersed over the entire recovery. The modeling analysis of Freund and Zouloumian (12) indicates a slightly more rapid recovery of muscle [L^-] relative to blood [L^-], although the assumptions of lactate release rate from the recovering muscles bearing a simple proportionality to muscle [L^-] (suggestive of a unitary process) and of the muscle itself operating as an homogenous structure are known to be oversimplifications. We are confined to noting that recovery of blood [L^-] (interpolated $t_{1/2}$ = 1,366 ± 799 s) was substantially slower than recovery of W′ and that there was no clear proportionality in the magnitude of [L^-] recovery relative to W′ recovery (Fig. 6*B*). This suggests that lactate recovery (and possibly glycogen repletion) is ongoing when W′ is fully recovered, consistent with earlier reports of slow blood [L^-] recovery kinetics (e.g., 12, 15, 35).

We can only speculate on the extent to which the recovery profile of intramuscular [L^-] might resemble that of blood [L^-] under the highly non-steady-state conditions of the present study. However, any involvement of intramuscular L^- clearance in W′ recovery kinetics is likely to be complex, including, for example, *1*) clearance to the interstitium via sarcolemmal monocarboxylate transporter (MCT) 1 and MCT4 action (e.g., 5, 6, 30), free diffusion, and nonspecific anion exchange (reviewed in Refs. 15, 35), *2*) subsequent L^- uptake and oxidation into adjacent oxidative fibers via MTC1-mediated transport (e.g., 19; cf. Ref. 63) and diffusion (reviewed in Refs. 6, 15, 35). Even so, simply on kinetic grounds, it seems unlikely that intramuscular L^- clearance is an exclusive mediator of W′ restoration.

Establishing the profile of intramuscular [H^+] is even more problematic. A considerable proportion (~70%) of H^+ clearance takes place in an MCT1- and MTC4-mediated 1:1 coupling with L^- (e.g., 5, 6, 30). Other routes include *1*) the influence of intramuscular P_{CO_2}, for which an appreciable transient increase has been reported early in recovery (e.g., 32), *2*) intramuscular H^+ production related to the rapid rates of PCr resynthesis early in recovery following the exhaustive conditioning bout (33), *3*) sarcolemmal Na^+/H^+-mediated H^+ efflux (reviewed in Ref. 30), and *4*) bicarbonate-mediated H^+ buffering in muscle and blood (with carbonic anhydrase exerting an important facilitating action on reaction rate) (3).

Substrate Depletion and Metabolite Accumulation

As discussed above, W′ is traditionally suggested to represent the depletion of putative substrate pools (e.g., 23, 42). We hypothesized that determining the recovery kinetics of such pools, such as might be reflected in the temporal profiles of $\dot{V}O_2$ as an index of intramuscular PCr kinetics (e.g., 48), for

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

example, might provide pointers elucidating the constitution of W′. Our data suggest, however, that a model of W′ as a "depletable" energy pool is likely to be too simple. Rather, the complex recovery kinetics of W′ suggest that it might be better reflected in the integrated action of variables that contribute to fatigue via the accumulation of key fatigue-inducing metabolites, such as *1*) inorganic phosphate through its effects on sarcoplasmic reticulum Ca^{2+} handling (e.g., 1), *2*) increased extracellular K^+ concentration, which predisposes to sarcolemmal depolarization and compromised membrane excitability (54), and *3*) the possibility of intramuscular oxidative stress (reviewed in Ref. 56), or *4*) an interaction of these effects. Our results suggest that subsequent studies might therefore be focused on identifying specific fatigue-inducing metabolites that have been postulated to represent W′, via procedures such as the serial sampling of muscle tissue by biopsy and of muscle-venous blood, as well as magnetic resonance spectroscopy of muscle (e.g., 17, 18, 48, 52).

Assumptions

While W′ is a mathematical construct implicit in the hyperbolic P-t_{LIM} relationship (14, 42, 45), it is one that compels physiological elucidation. A fundamental assumption in the present study was that the supra-CP conditioning bout fully depleted W′ at the tolerable limit or that its mediator(s) is depleted to a low, but limiting, value. Furthermore, as we are aware of no technique that is available for direct assessment of the time course of W′ recovery following fatiguing exercise, it was also assumed that the W′ estimate obtained from each of the postconditioning P-t_{LIM} relationships reflected the value to which W′ would have recovered by the end of the intervening recovery period; i.e., it was assumed that there was no further recovery during the postconditioning bout itself. This assumption is supported by the demonstration that no significant amount of work can be performed at a lower WR within the very heavy-intensity domain (i.e., supra-CP) immediately following attainment of the tolerable limit, suggesting that the "flux" of W′ is unidirectional above CP (7, 43).

We also cannot rule out the possibility that the recovery profile of W′ following exhaustive exercise may have been modified as a result of an altered work efficiency effect (52) on the W′ depletion rate. However, currently available techniques do not allow for a more direct estimation of the rate of W′ utilization during volitional exercise.

Conclusions

The retained hyperbolic P-t_{LIM} relationship, with unchanged CP, following fatiguing exercise supports the assertion that W′ depletion shapes high-intensity exercise tolerance. We suggest that the complexity of its recovery kinetics may be more consistent with W′ reflecting the accumulation of fatigue-related metabolites, rather than expenditure of a simple source of stored energy. However, it is also possible that an integrated function of these processes is involved in determining W′. Resolution of these issues will require more invasive targeting of particular mechanisms.

ACKNOWLEDGMENTS

The authors thank Brendan McDermott for assistance during data collection.

DISCLOSURES

No conflicts of interest are declared by the author(s).

REFERENCES

1. **Allen DG, Lamb GD, Westerblad H.** Impaired calcium release during fatigue. *J Appl Physiol* 104: 296–305, 2008.
2. **Barstow TJ, Lamarra N, Whipp BJ.** Modulation of muscle and pulmonary O_2 uptakes by circulatory dynamics during exercise. *J Appl Physiol* 68: 979–989, 1990.
3. **Beaver WL, Wasserman K, Whipp BJ.** Bicarbonate buffering of lactic acid generated during exercise. *J Appl Physiol* 60: 472–478, 1986.
4. **Beaver WL, Wasserman K, Whipp BJ.** On-line computer analysis and breath-by-breath graphical display of exercise function tests. *J Appl Physiol* 34: 128–132, 1973.
5. **Bonen A.** The expression of lactate transporters (MCT1 and MCT4) in heart and muscle. *Eur J Appl Physiol* 86: 6–11, 2001.
6. **Brooks GA.** Intra- and extra-cellular lactate shuttles. *Med Sci Sports Exerc* 32: 790–799, 2000.
7. **Coats EM, Rossiter HB, Day JR, Miura A, Fukuba Y, Whipp BJ.** Intensity-dependent tolerance to exercise after attaining $\dot{V}O_{2\,max}$ in humans. *J Appl Physiol* 95: 483–490, 2003.
8. **Day JR, Rossiter HB, Coats EM, Skasick A, Whipp BJ.** The maximally attainable $\dot{V}O_2$ during exercise in humans: the peak vs. maximum issue. *J Appl Physiol* 95: 1901–1907, 2003.
9. **Favero TG, Zable AC, Colter D, Abramson JJ.** Lactate inhibits Ca^{2+}-activated Ca^{2+} channel activity from skeletal muscle sarcoplasmic reticulum. *J Appl Physiol* 82: 447–452, 1997.
10. **Ferguson C, Whipp BJ, Cathcart AJ, Rossiter HB, Turner AP, Ward SA.** Effects of prior very-heavy intensity exercise on indices of aerobic function and high-intensity exercise tolerance. *J Appl Physiol* 103: 812–822, 2007.
11. **Fitts RH.** The cross-bridge cycle and skeletal muscle fatigue. *J Appl Physiol* 104: 551–558, 2008.
12. **Freund H, Zouloumian P.** Lactate after exercise in man. IV. Physiological observations and model predictions. *Eur J Appl Physiol Occup Physiol* 46: 161–176, 1981.
13. **Fukuba Y, Miura A, Endo M, Kan A, Yanagawa K, Whipp BJ.** The curvature constant parameter of the power-duration curve for varied-power exercise. *Med Sci Sports Exerc* 35: 1413–1418, 2003.
14. **Fukuba Y, Whipp BJ.** A metabolic limit on the ability to make up for lost time in endurance events. *J Appl Physiol* 87: 853–861, 1999.
15. **Gladden LB.** Lactate metabolism: a new paradigm for the third millennium. *J Physiol* 558: 5–30, 2004.
16. **Gollnick PD, Armstrong RB, Sembrowich WL, Shepherd RE, Saltin B.** Glycogen depletion pattern in human skeletal muscle fibers after heavy exercise. *J Appl Physiol* 34: 615–618, 1973.
17. **Grassi B, Poole DC, Richardson RS, Knight DR, Erickson BK, Wagner PD.** Muscle O_2 uptake kinetics in humans: implications for metabolic control. *J Appl Physiol* 80: 988–998, 1996.
18. **Harris RC, Edwards RH, Hultman E, Nordesjo LO, Nylind B, Sahlin K.** The time course of phosphorylcreatine resynthesis during recovery of the quadriceps muscle in man. *Pflügers Arch* 367: 137–142, 1976.
19. **Hashimoto T, Hussien R, Brooks GA.** Colocalization of MCT1, CD147, and LDH in mitochondrial inner membrane of L6 muscle cells: evidence of a mitochondrial lactate oxidation complex. *Am J Physiol Endocrinol Metab* 290: E1237–E1244, 2006.
20. **Heigenhauser GJ, Sutton JR, Jones NL.** Effect of glycogen depletion on the ventilatory response to exercise. *J Appl Physiol* 54: 470–474, 1983.
21. **Hermansen L, Vaage O.** Lactate disappearance and glycogen synthesis in human muscle after maximal exercise. *Am J Physiol Endocrinol Metab Gastrointest Physiol* 233: E422–E429, 1977.
22. **Heubert RA, Billat VL, Chassaing P, Bocquet V, Morton RH, Koralsztein JP, di Prampero PE.** Effect of a previous sprint on the parameters of the work-time to exhaustion relationship in high intensity cycling. *Int J Sports Med* 26: 583–592, 2005.
23. **Hill DW.** The critical power concept. *Sports Med* 16: 237–254, 1993.
24. **Hill DW, Smith JC.** A comparison of methods of estimating anaerobic work capacity. *Ergonomics* 36: 1495–1500, 1993.
25. **Hogan MC, Gladden LB, Kurdak SS, Poole DC.** Increased [lactate] in working dog muscle reduces tension development independent of pH. *Med Sci Sports Exerc* 27: 371–377, 1995.

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.

26. **Housh TJ, Devries HA, Housh DJ, Tichy MW, Smyth KD, Tichy AM.** The relationship between critical power and the onset of blood lactate accumulation. *J Sports Med Phys Fitness* 31: 31–36, 1991.
27. **Hughson RL, Orok CJ, Staudt LE.** A high velocity treadmill running test to assess endurance running potential. *Int J Sports Med* 5: 23–25, 1984.
28. **Jones AM, Wilkerson DP, Burnley M, Koppo K.** Prior heavy exercise enhances performance during subsequent perimaximal exercise. *Med Sci Sports Exerc* 35: 2085–2092, 2003.
29. **Jones AM, Wilkerson DP, DiMenna F, Fulford J, Poole DC.** Muscle metabolic responses to exercise above and below the "critical power" assessed using ^{31}P-MRS. *Am J Physiol Regul Integr Comp Physiol* 294: R585–R593, 2008.
30. **Juel C.** Lactate-proton cotransport in skeletal muscle. *Physiol Rev* 77: 321–358, 1997.
31. **Karlsson J, Saltin B.** Lactate, ATP and CP in working muscles during exhaustive exercise in man. *J Appl Physiol* 29: 598–602, 1970.
32. **Kowalchuk JM, Heigenhauser GJ, Lindinger MI, Sutton JR, Jones NL.** Factors influencing hydrogen ion concentration in muscle after intense exercise. *J Appl Physiol* 65: 2080–2089, 1988.
33. **Kushmerick MJ.** Multiple equilibria of cations with metabolites in muscle bioenergetics. *Am J Physiol Cell Physiol* 272: C1739–C1747, 1997.
34. **Lamarra N, Whipp BJ, Ward SA, Wasserman K.** Effect of interbreath fluctuations on characterizing exercise gas exchange kinetics. *J Appl Physiol* 62: 2003–2012, 1987.
35. **Lindinger MI, McKelvie RS, Heigenhauser GJ.** K^+ and Lac^- distribution in humans during and after high-intensity exercise: role in muscle fatigue attenuation? *J Appl Physiol* 78: 765–777, 1995.
36. **McLellan TM, Cheung KS.** A comparative evaluation of the individual anaerobic threshold and the critical power. *Med Sci Sports Exerc* 24: 543–550, 1992.
37. **Messer JI, Jackman MR, Willis WT.** Pyruvate and citric acid cycle carbon requirements in isolated skeletal muscle mitochondria. *Am J Physiol Cell Physiol* 286: C565–C572, 2004.
38. **Miura A, Kino F, Kajitani S, Sato H, Fukuba Y.** The effect of oral creatine supplementation on the curvature constant parameter of the power-duration curve for cycle ergometry in humans. *Jpn J Physiol* 49: 169–174, 1999.
39. **Miura A, Sato H, Whipp BJ, Fukuba Y.** The effect of glycogen depletion on the curvature constant parameter of the power-duration curve for cycle ergometry. *Ergonomics* 43: 133–141, 2000.
40. **Monod H.** How muscles are used in the body. In: *The Structure and Function of Muscle. Structure,* edited by Bourne GH. New York: Academic, 1972, vol. 1, pt. I, p. 23–74.
41. **Monod H, Scherrer J.** The work capacity of a synergic muscular group. *Ergonomics* 8: 329–338, 1965.
42. **Moritani T, Nagata A, deVries HA, Muro M.** Critical power as a measure of physical work capacity and anaerobic threshold. *Ergonomics* 24: 339–350, 1981.
43. **Morton RH.** The critical power and related whole-body bioenergetic models. *Eur J Appl Physiol* 96: 339–354, 2006.
44. **Özyener F, Rossiter HB, Ward SA, Whipp BJ.** Influence of exercise intensity on the on- and off-transient kinetics of pulmonary oxygen uptake in humans. *J Physiol* 533: 891–902, 2001.
45. **Poole DC, Ward SA, Gardner GW, Whipp BJ.** Metabolic and respiratory profile of the upper limit for prolonged exercise in man. *Ergonomics* 31: 1265–1279, 1988.
46. **Poole DC, Ward SA, Whipp BJ.** The effects of training on the metabolic and respiratory profile of high-intensity cycle ergometer exercise. *Eur J Appl Physiol Occup Physiol* 59: 421–429, 1990.
47. **Pringle JS, Jones AM.** Maximal lactate steady state, critical power and EMG during cycling. *Eur J Appl Physiol* 88: 214–226, 2002.
48. **Rossiter HB, Ward SA, Kowalchuk JM, Howe FA, Griffiths JR, Whipp BJ.** Dynamic asymmetry of phosphocreatine concentration and O_2 uptake between the on- and off-transients of moderate- and high-intensity exercise in humans. *J Physiol* 541: 991–1002, 2002.
49. **Rossiter HB, Whipp BJ, Ward SA, McIntyre DJ, Griffiths JR, Howe FA.** The heterogeneity of intramuscular metabolism using simultaneous ^{31}P 2D-CSI and pulmonary oxygen uptake ($\dot{V}O_2$) during incremental knee-extensor exercise in humans (Abstract). *Proc Int Soc For Magn Reson Med (Kyoto)* 12: 773, 2004.
50. **Sahlin K, Alvestrand A, Brandt R, Hultman E.** Intracellular pH and bicarbonate concentration in human muscle during recovery from exercise. *J Appl Physiol* 45: 474–480, 1978.
51. **Sahlin K, Katz A, Broberg S.** Tricarboxylic acid cycle intermediates in human muscle during prolonged exercise. *Am J Physiol Cell Physiol* 259: C834–C841, 1990.
52. **Sahlin K, Sorensen JB, Gladden LB, Rossiter HB, Pedersen PK.** Prior heavy exercise eliminates $\dot{V}O_2$ slow component and reduces efficiency during submaximal exercise in humans. *J Physiol* 564: 765–773, 2005.
53. **Saltin B, Karlsson J.** Muscle glycogen utilization during work of different intensities. In: *Muscle Metabolism During Exercise*, edited by Pernow B and Saltin B. New York: Plenum, 1972, p. 289–299.
54. **Sjogaard G.** Exercise-induced muscle fatigue: the significance of potassium. *Acta Physiol Scand Suppl* 593: 1–63, 1990.
55. **Smith JC, Stephens DP, Hall EL, Jackson AW, Earnest CP.** Effect of oral creatine ingestion on parameters of the work rate-time relationship and time to exhaustion in high-intensity cycling. *Eur J Appl Physiol Occup Physiol* 77: 360–365, 1998.
56. **Smith MA, Reid MB.** Redox modulation of contractile function in respiratory and limb skeletal muscle. *Respir Physiol Neurobiol* 151: 229–241, 2006.
57. **Thomson JA, Green HJ, Houston ME.** Muscle glycogen depletion patterns in fast twitch fiber subgroups of man during submaximal and supramaximal exercise. *Pflügers Arch* 379: 105–108, 1979.
58. **Vanhatalo A, Burnley M.** Influence of prior sprint exercise on the parameters of the "all-out critical power test" in men. *Exp Physiol* 94: 255–263, 2009.
59. **Whipp BJ.** Domains of aerobic function and their limiting parameters. In: *The Physiology and Pathophysiology of Exercise Tolerance*, edited by Steinacker JM and Ward SA. New York: Plenum, 1996, p. 83–89.
60. **Whipp BJ, Huntsman DJ, Storer T, Lamarra N, Wasserman K.** A constant which determines the duration of tolerance to high-intensity work (Abstract). *Fed Proc* 41: 1591, 1982.
61. **Whipp BJ, Özyener F.** The kinetics of exertional O_2 uptake: assumptions and inferences. *Med Sport* 51: 39–49, 1998.
62. **Whipp BJ, Ward SA, Wasserman K.** Respiratory markers of the anaerobic threshold. *Adv Cardiol* 35: 47–64, 1986.
63. **Yoshida Y, Holloway GP, Ljubicic V, Hatta H, Spriet LL, Hood DA, Bonen A.** Negligible direct lactate oxidation in subsarcolemmal and intermyofibrillar mitochondria obtained from red and white rat skeletal muscle. *J Physiol* 582: 1317–1335, 2007.

Downloaded from journals.physiology.org/journal/jappl (157.157.039.214) on January 7, 2021.